Lab 1: Preprocessing & Data Wrangling

Course: **INS-605: Data Analysis**
Lecturer: **Sothea HAS, PhD**

Objective: In this lab, we will work with real-world product review data (`Amazon Product Reviews`) to practice key data wrangling techniques discussed in the lecture: inspecting raw data, handling missing values, converting data types, feature transformation, grouping/aggregation, combining datasets, and reshaping data.

> The notebook of the lab can be downloaded from [Lab1: Proprocessing & Data Wrangling].

- Student name: ...
- ID: ...

> **Submission:** You must do the followings
>
> - Name your file as: `Lab1_name.ipynb`
> - Submit this jupyter notebook file (`Lab1_name.ipynb`) to Canvas, NOT THE COLAB LINK!

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1. Importing and Inspecting the Data

We will explore the **Amazon Product Reviews** dataset from Kaggle (`arhamrumi/amazon-product-reviews`). Let's download and load the dataset into pandas.

▼ Code

```
# %pip install kagglehub

import kagglehub
import pandas as pd
import os

# Download latest version
path = kagglehub.dataset_download("arhamrumi/amazon-product-rev

# Load data (locate the CSV file in the downloaded path)
file_name = [f for f in os.listdir(path) if f.endswith('.csv')]
```

```python
data = pd.read_csv(f"{path}/{file_name}")
data.head()
```

1.1. Overview of the data

Before cleaning, inspect the structure of the dataset and answer the following:

A. Check the dimension of the dataset (`shape`). How many rows and columns are there?

B. What are the data types of each column (`dtypes`)? Identify columns whose data types do not match their intended semantic meaning.

C. Check for missing values (`isna().sum()`). Which columns contain `NaN` values, and how many are missing in each column?

D. Are there any duplicated rows in this dataset? Calculate the total number of duplicates using `.duplicated().sum()`.

E. Check the number of unique values in each column (`nunique()`).

- **E.1.** Compare the number of unique `UserId`s to unique `ProfileName`s. Why do you think we observed such numbers?
- **E.2.** Do missing names in `ProfileName` potentially share a `UserId` with non-missing rows? This will guide our imputation strategy.

F. Check if missing `ProfileName`s are related to missing `Summary`s.

```python
# Let's find out!
```

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2. Data Cleaning

Now that you have identified the issues in raw data, let's clean them step-by-step.

F. Datetime Conversion: Convert the `Time` column (Unix timestamp in seconds) into a proper pandas `datetime` format using `pd.to_datetime(..., unit='s')`.

G. Handling Missing Text Values (Smart Imputation): - **G1. Recovering Profile Names:** Since users might have multiple reviews, a missing `ProfileName` might have a known `ProfileName` under the same `UserId` in another row. Check if there are such cases, therefore use this information to impute the missing values. - **G2. Fallback Imputation:** Fill any *remaining* missing values in `ProfileName` with `'Unknown'`, and fill missing values in `Summary` with `'No Summary'`.

H. Duplicates: If any duplicated rows exist, drop them using `.drop_duplicates()` and verify the shape of your DataFrame afterward.

```
# To do
```

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3. Feature Transformation

Feature engineering helps turn raw fields into more useful analytical features.

I. Rating Classification (`Status`): Create a new column named `Status` based on the review rating (`Score`):

- `'High'` if `Score >= 4`
- `'Medium'` if `Score == 3`
- `'Low'` if `Score <= 2`

J. Temporal Extraction (`Year`): Extract the year from your converted `Time` column and store it in a new column called `Year`.

K. Metric Calculation (`HelpfulnessRatio`): Create a column `HelpfulnessRatio` by dividing `HelpfulnessNumerator` by `HelpfulnessDenominator`. Handle division-by-zero cases (e.g., set to 0 when `HelpfulnessDenominator == 0`).

```
# To do
```

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4. Grouping & Aggregation

L. Most Reviewed Products: Group by `ProductId`. Find the **top 5 products** with the highest number of reviews.

M. User Activity: Group by `UserId`. Find the **top 5 most active reviewers** (users with the most reviews) and calculate their average review `Score`.

N. Product Metrics: For the top 5 products identified in Question **L**, use `.groupby()` and `.agg()` to compute both the average `Score` and total count of reviews.

```
# To do
```

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5. Combining and Reshaping Data

O. Combining Datasets (`pd.merge`): Create a small lookup DataFrame named `product_catalog` containing:

- `ProductId`: The top 5 product IDs from Question **L**
- `Category`: Assigned categories (e.g., `'Beverages'`, `'Snacks'`, `'Gourmet'`, `'Pet Supplies'`, `'Organic'`)

Perform an **inner join** (`pd.merge`) between your aggregated product metrics from Question **N** and `product_catalog`.

P. Reshaping Data (`pd.pivot_table`): Create a pivot table using `pd.pivot_table()` to summarize review volume:

- **Index (Rows):** `Year`
- **Columns:** `Status` (`High`, `Medium`, `Low`)
- **Values:** Count of reviews (`Score`)
- **Mean:** average score per category of `Status`.
- Fill missing cells with 0.

Q. How does the platform perform over the years?

```
# To do
```

Further Reading

- `Pandas` Data Wrangling Cheat Sheet: https://pandas.pydata.org/Pandas_Cheat_Sheet.pdf

- Chapter 6-8, Python for Data Analysis by Wes McKinney: https://wesmckinney.com/book/

- `Pandas` Apply, Transform, Map: https://pandas.pydata.org/docs/user_guide/basics.html